

Environmental and Demographic Influences on PM2.5 Levels

Lloyd D'Silva
64858

fc64858@alunos.ciencias.ulisboa.pt

Matei-Alexandru Lupaşcu
64471

fc64471@alunos.ciencias.ulisboa.pt

Yubing Chen
52915

fc52915@alunos.ciencias.ulisboa.pt

ABSTRACT

This project investigates the environmental and demographic factors influencing PM2.5 levels, a critical indicator of air pollution. Our main research questions address identifying key environmental and demographic factors, quantifying urbanization's impact on PM2.5, determining if capital cities tend to be more polluted, analyzing the influence of seasonal changes, and assessing the impact of deforestation. To address these, we integrated and analyzed five datasets: Global Weather Repository, Traffic List for Country, App Air Quality Database 2018, Forest Coverage List, and Global PM2.5 (1998–2022). Our methodology involved data integration, identity resolution using multiple similarity metrics, rigorous data cleaning, and the construction of a data mart in SQLite. The report's conclusion will present the findings of our analysis, and all annexes, along with the reports on the three individual challenges, are provided at the end of this document. All the project files are available in our Github repository. <https://github.com/iceyubing/ipai25/tree/phase-2>

KEYWORDS

PM2.5, Particulate Matters, Pollution, Environmental Factors, Demographics, Forest Coverage, Air Quality, Traffic.

1. INTRODUCTION

PM2.5 refers to fine particulate matter with a diameter less than or equal to 2.5 micrometers. In environmental science, it is considered one of the main indicators of air pollution, as it represents the excessive dispersion and suspension of particles in the air caused by human activities. In recent years, air pollution has become increasingly associated with public health, making PM2.5 a highly discussed and relevant topic.

For this topic, our project focuses on research analyzing which factors influence the PM2.5 index. Based on this core question, we have established several hypotheses, including: the number of vehicles per capita in a country, the population size, the percentage of forest coverage, and local weather parameters. We will analyze data to examine whether these factors have an impact on PM2.5 levels. Additionally, we will compare the national average PM2.5 index with the PM2.5 index of capital cities to investigate whether the index tends to be higher in capitals.

This first phase of the project aims to find one (or several) data sources that can further be passed through an entity matching process.

In this project, we are considering five datasets: **Global Weather Repository**, **Traffic List for Country**, **App Air Quality**

Database 2018, **Forest Coverage List** and **Global PM2.5 (1998–2022)**. (Annex 1)

Global Weather Repository provides daily weather information for worldwide locations and for an exact period, such as temperature, precipitation, air quality, among others, with data sourced from Kaggle.

The Traffic List for Country provides data on the number of vehicles per capita for each city, sourced from Wikipedia.

The App Air Quality Database 2018 provides PM2.5 index data for various cities in each country, sourced from WHO Database.

The Forest Coverage List contains information about forest coverage and population for each country, with data sourced from Kaggle.

The Global PM2.5 (1998–2022) dataset provides the average PM2.5 levels for each country, sourced from Atmospheric Composition Analysis Group, Washington University in St. Louis, USA.

In this report, we have presented and analyzed the attributes and values of each dataset, using Jupyter Notebook, Excel and SQLite.

2. INTEGRATION AND ANALYTIC DATA PROCESSING

First, we broke down each attribute from all previously collected datasets into more detailed, smaller tables, and clearly annotated the meaning of each attribute (see Annex 3 for details). Then, we re-integrated the data based on similar attributes and restructured the schema accordingly (the result is shown in Annex 5).

In this step, we organized and integrated similar data from different sources into unified dimension tables. For example, the **Location** table consolidates geographical information from the *Global Weather*, *Forest Cover Rate*, and *Global Air Quality - Country* datasets. Similarly, the **Population_Coverage** dimension combines population-related data from both the *Forest Cover Rate* and *Global Air Quality - Country* sources. (see the annex 4)

Furthermore, we integrate attributes from different data sources when they convey the same underlying meaning. In certain instances, different authors may adopt varying but equivalent terminology to describe the same attribute for distinct analytical purposes (see Annex 1).

3. IDENTITY RESOLUTION

Blocking strategy:

To efficiently compare and link records across multiple datasets (forest, air_city, air_country, weather, and traffic), we implemented a prefix-based blocking strategy. For both city and country attributes, we extracted the first three characters from normalized versions of these fields, creating standardized prefixes (e.g., "Portugal" → "por"). As suggested in the Peer Review recommendations, one alternative for the blocking strategy could have been by using the ISO code for countries. The reason why we did not implement it using that strategy consists in the fact that ISO codes are not available for the city data, therefore we preferred a more general approach that can apply to both country and city names. This approach reduces the comparison space by grouping records with identical prefixes before applying more computationally expensive similarity measures. We then computed intersections of these prefix sets across datasets to identify potential matches. This method ensures scalability while maintaining recall, as it focuses on subsets of data with high initial overlap.

In the identity resolution phase of our project, we implemented a rule-based approach to match and link entities across different datasets. This approach leverages three distinct **similarity metrics** to ensure accurate and robust matching of both **country** and **city** names. Each metric was chosen based on its strengths in handling specific types of variations in textual data.

3.1 Jaro-Winkler similarity (character-based)

Description:

Jaro-Winkler measures similarity between two strings by comparing characters and their order, with a bonus for matching prefixes. It is highly effective for detecting minor spelling variations, abbreviations, or transpositions (e.g. "Algeria" vs. "Algerie").

Application:

We applied Jaro-Winkler to match country names across datasets where minor spelling differences or abbreviations were present. A threshold of 0.90 was set to ensure high confidence in matches.

Why did we use it:

This metric excels in cases where strings are similar but not identical, making it ideal for datasets with inconsistent formatting or truncations.

3.2 Levenshtein distance (edit distance-based)

Description:

Measures the minimum number of single-character edits (insertions, deletions, substitutions) needed to transform one string into another.

Application:

We set a max edit distance of 1 to catch typos or close variants. We initially established a threshold of 2, but we inspected the matched data and observed that many country names only differ by a small number of characters. Therefore, the initial threshold did not only include slightly misspelled names, but totally different countries as well.

Why did we use it:

Perfect for rigid, small-scale discrepancies where even a single character difference should be flagged.

3.3 Cosine similarity with TF-IDF (token-based)

Description:

A token-based metric that treats strings as vectors of word tokens (using TF-IDF weighting) and computes the cosine of the angle between them. This captures semantic similarity beyond character-level differences.

Application:

Used for multi-word names (e.g., "Sao Tome and Principe") where word order or minor term variations exist. A threshold of 0.90 ensured high confidence matches.

Why did we use it:

Best for compound names or phrases where meaning matters more than exact spelling (e.g., "United Kingdom" vs. "UK" after normalization).

3.4 Implementation workflow

Normalization: Lowercasing, removing accents/special characters.

Threshold Tuning: Balanced precision/recall (e.g., Jaro-Winkler's 0.90 threshold avoids false matches like "Iran" vs. "Iraq").

Cross-Validation: Matches confirmed by multiple metrics, based on inspecting the matched data between the different sources.

This hybrid approach ensured robust matching across datasets with diverse naming conventions.

4. DATA CLEANING

The *dp.py* file handles data cleaning for all datasets. While the *traffic*, *air_city*, and *air_country* datasets had no missing values according to Pandas Profiling, the forest dataset contained several gaps. These were addressed using specific methods implemented in *dp.py*, as shown in the Jupyter notebook, which also includes an example from row 175. To improve data quality, large datasets were filtered to remove irrelevant information, with results detailed in Annex 2. Normalization was also performed – standardizing country and city names to lowercase and removing special characters – to support accurate matching.

For the individual challenge 64858, a simulated messy dataset called GlobalWeatherRepository was used to showcase a complete data cleaning workflow. This challenge mimicked real-world issues by introducing deliberate errors and inconsistencies, such as formatting issues and incorrect values. The goal was to demonstrate how these problems would be handled using a structured, multi-step cleaning pipeline, ensuring high data quality through advanced techniques.

The cleaning process involved detecting hidden missing values (like "-" and "N/A"), correcting data types and formats with regular expressions, and inputting or removing flawed data. Extreme outliers and illogical values were addressed using domain knowledge and IQR analysis. Duplicate records, including near matches like "NYC" vs. "New York City," were resolved using fuzzy matching. Text normalization, date validation, and the removal of irrelevant or fake entries like "Test City" ensured the final dataset was clean, consistent, and suitable for reliable analysis.

5. DATA PROFILING AND SOLUTION

Inside the notebook we applied Pandas Profiling for each dataset, after cleaning the columns beforehand. It was observed that the

traffic, *air_city*, *weather* and *air_country* datasets do not contain any missing values. However, missing values were identified in the forest dataset. The resulted HTML file highlights the missing values in the forest dataset and details the methods applied to address them, which were implemented in the *dp.py* file.

Therefore, we inspected the missing values and decided to eliminate some rows where data was missing from all the columns corresponding to the forest percentage in each year. For one of the missing values, we decided to substitute the null value with the average of the other ones within the same row.

6. DATA FUSION

Actual Scenarios

1. Inconsistent Country Names

Strategy Used: *Trust your friends*

To address inconsistencies in country names (e.g., "USA" vs "United States"), we applied the *Trust your friends* strategy by adopting a standardized naming convention based on a trusted authority, such as the ISO or World Bank. When conflicting variants were found, we retained the version used by the most reliable source. A mapping table was constructed (*country_mapping.csv*) to ensure consistency across all datasets.

2. Missing Country Data

Strategy Used: *Pass it on*

In the 52915 personal challenge dataset, certain anomalies were identified – for example, countries like the United States and other Western nations did not report GDP data for specific years, resulting in values of zero. To ensure factual accuracy and data integrity, we chose to remove such records. This approach aligns with the *Pass it on* strategy, which escalates unresolved conflicts by ignoring unreliable or insufficient data.

3. Missing Population Data

Strategy Used: *Take the information*

Two population-related datasets were used in this report. However, due to differences in source and data collection periods, the forest-cover-v1.csv dataset only included population data for the year 2020, while the GlobalPM25-1998-2022.csv dataset lacked this information entirely. Given the structural requirements of the data mart, we applied the available 2020 population data to fill in the missing values for other years. This follows the *Take the information* strategy, classified as Instance-based, Avoiding, where non-null values are preferred over missing data.

Hypothetical Scenarios [There are examples code in notebook]

4. Population Data Conflict

Strategy Used: *Keep up to date*

Suppose two data sources provide conflicting population values for the same country and year:

- Source A (official, published 2020): 139 million
- Source B (third-party, published 2020): 138 million

In such cases, we would apply the *Keep up to date* strategy, classified as Metadata-based, Deciding, by selecting the most recently published or updated value, assuming it reflects the most accurate information.

5. Forest Coverage Conflict

Strategy Used: *Meet in the middle*

Consider three different sources reporting varying forest coverage values for the same country and year:

- Source A: 65.2% [2020]
- Source B: 64.8% [2020]

We would use the *Meet in the middle* strategy, classified as Instance-based, Mediating, to compute the average value ($\approx 65\%$). This approach ensures a fair and neutral resolution of numerical conflicts.

6. Air Quality Data Conflict

Strategy Used: *Trust your friends*

When PM2.5 concentration data varies across sources with different levels of credibility:

- Source A (satellite-based estimation): 32.5 $\mu\text{g}/\text{m}^3$
- Source B (ground-based monitoring station): 28.1 $\mu\text{g}/\text{m}^3$
- Source C (academic model): 30.2 $\mu\text{g}/\text{m}^3$

We apply a weighted average based on source reliability – assigning higher weight to ground-based data (e.g., 0.5), followed by satellite (0.3), and academic model (0.2). This reflects the *Trust your friends* strategy, classified as Metadata-based, Deciding, where data from more credible sources is prioritized in the fusion process.

7. DATA MARTS AND ANALYSIS

7.1 Building the Data Mart - SQLite

Our data mart is built upon an SQLite3-based data warehouse. During the data preparation phase, we restructured the existing datasets to fit the warehouse architecture. Specifically, we separated the year attribute from other data fields, organizing it as an independent attribute to support a more normalized schema.

Subsequently, we constructed the data architecture (as illustrated in Annex 6) based on a fact table (PM2.5Regist) and multiple related dimension tables, following the star schema model. The data was integrated and loaded using SQLite3 Studio. (Due to size limitations, the PM2.5-city and GlobalWeatherRepository datasets were not imported into the database. Instead, they were processed separately as CSV files using Pandas.)

The data mart supports environmental, demographic, and socio-economic analysis across countries and years, enabling multi-dimensional insights into sustainability, public health, and development patterns for further research.

We compiled a list of example questions that can be answered using the data in the warehouse (Annex 7).

7.2 Data Analysis

The initial step involved loading several data tables from an SQLite database, *ipai25.db*, which serves as the project's data mart. A custom Python function, *load_table_from_db*, was developed to facilitate this process.

The following tables were successfully loaded:

- **Country_information:** Contains static information about countries, including Country, Country Code, Capital, Continent, and Area.
- **PM2.5-Country:** Provides PM2.5 concentration data per country and year.
- **Forest_coverage:** Contains data on forest area coverage by country and year.
- **Trafficlist:** Includes information on vehicle traffic rates and numbers per country and year.
- **Population_coverage:** Details population statistics, including total population, density, and coverage percentages.

A critical check was implemented post-loading to ensure that all essential dataframes were populated, preventing further analysis if any critical data was missing.

7.2.1 Data preprocessing and merging

Column Renaming and Standardization

Prior to merging, column names across all dataframes were standardized to ensure consistency and facilitate accurate joins. This involved stripping whitespace and newline characters and replacing multiple spaces with single spaces. [Specified in the Jupyter Notebook]

In-depth Cleaning of Merge Keys

A dedicated function was applied to the 'Country' and 'Year' columns across all relevant dataframes. This function performed stripping leading/trailing whitespace, replacing non-breaking spaces with standard spaces, normalizing Unicode characters to their closest ASCII representation to resolve encoding inconsistencies, and converting 'Year' columns to string type to ensure consistent merging behavior.

Merging Dataframes

The cleaned dataframes were merged sequentially into a single analytical dataframe, `df_analysis`, primarily using 'Country' and 'Year' as common keys. An inner join was used for `df_pm25_country` and `df_country_info` to retain only matching records, while left joins were used for other dataframes (`df_population_coverage`, `df_forest_coverage`, `df_traffic_list`) to preserve all PM2.5 records and append relevant information where available.

7.2.2 Debugging for Trafficlist Data

During the merging process, it was observed that the `Rate_per_1000` and `Vehicles` columns from the `Trafficlist` table contained a high number of NaN values in the merged `df_analysis`. A targeted debugging effort was undertaken to investigate this data sparsity.

Targeted Debugging Example (India, 2015)

To confirm the root cause, a specific example (India, 2015) was examined. Both `df_pm25_country` and `df_traffic_list` contained data for this specific country-year combination, and their 'Country' and 'Year' columns were confirmed to be of object (string) dtype with consistent raw values. This confirmed that the cleaning and type conversion steps were effective for these specific instances and that the NaN values were indeed due to a lack of corresponding data in the `Trafficlist` for many country-year combinations present in the PM2.5-Country data.

7.2.3 Analysis and Visualizations

Analysis 1 - The Influence of the Number of Vehicles on PM2.5

Despite an initial hypothesis of a positive correlation, the data revealed a moderate and statistically significant inverse relationship between the vehicle rate per 1000 people and PM2.5 concentrations (Pearson correlation: -0.41, p-value: 0.000). This suggests that as the vehicle rate increases, PM2.5 tends to decrease. However, it's important to note that the vehicle rate only explains about 17.2% of the variation in PM2.5 levels, indicating that many other factors are at play. This counter-intuitive finding could be attributed to data limitations, effective pollution mitigation strategies in countries with higher vehicle rates, or the more dominant influence of other environmental and industrial factors.

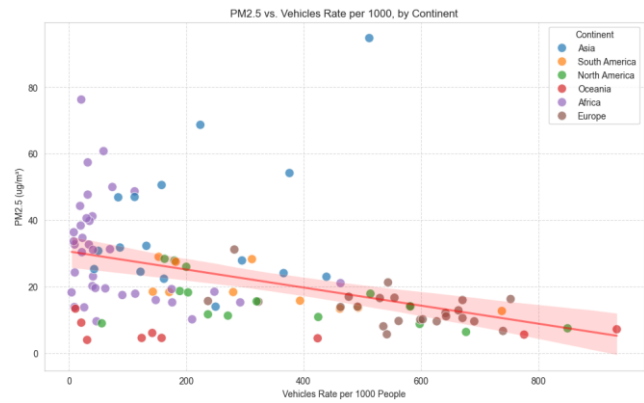


Figure 1. PM2.5 vs. Vehicles Rate per 1000

Analysis 2 - Influence of Population on PM2.5

For this analysis, the visualizations indicate a nuanced relationship between population and PM2.5. While the box plot suggests that higher population density generally correlates with increased PM2.5 levels, the scatter plots show that total population alone is not a straightforward predictor. The relationship is complex, with significant regional variations (e.g., Asia showing higher PM2.5 in large populations, similarly Oceania has lowest PM2.5 independently of the country's population) and other factors likely influencing air quality.

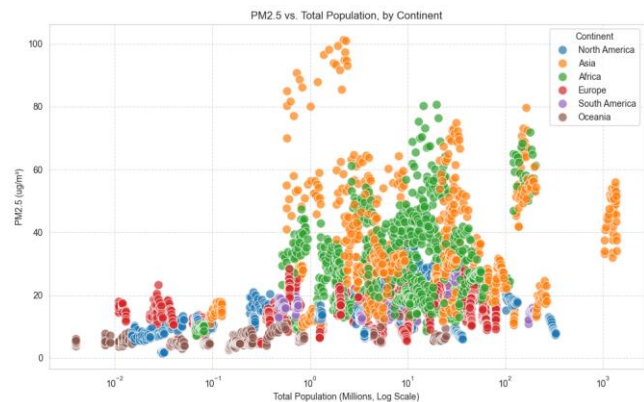


Figure 2. PM2.5 vs. Population

Analysis 3 - Influence of Forest coverage on PM2.5

For the third analysis, the visualizations suggest a moderately weak negative correlation between forest area percentage and PM2.5 concentration (Pearson correlation: -0.28, p-value: 0.000). The scatter plot shows a general tendency for PM2.5 to be slightly lower in countries with higher forest coverage, but the relationship is not very strong, and there is considerable scatter. While plotting the forest coverage area on top of the PM2.5 average values of the first five countries in our data mart, we have noticed that constant forest coverage areas could lead to both increases and decreases in the PM2.5 index, as other attributes contribute as well. However, in countries such as Albania or Algeria, where there is clear evidence of a forestation tendency in the last years, the values of PM2.5 have been moderately decreasing.

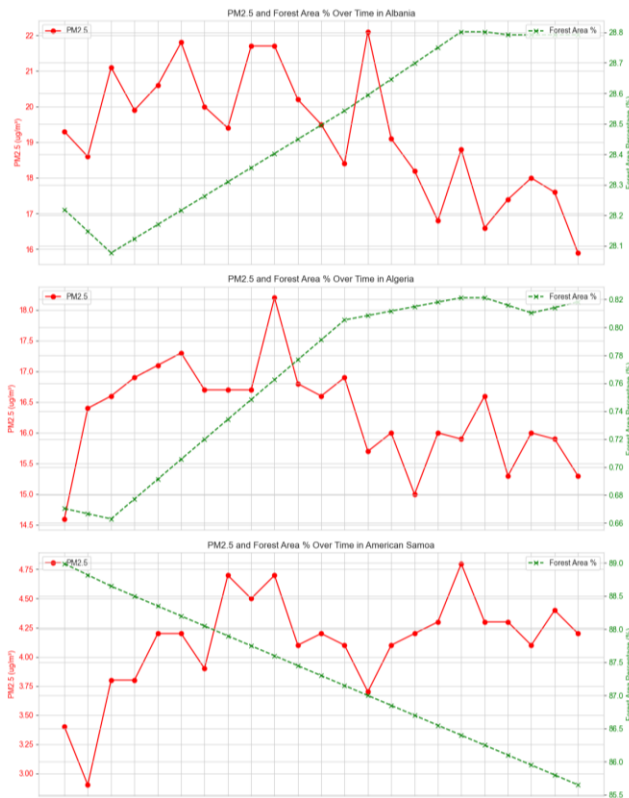


Figure 3. PM2.5 and Forest Coverage in three countries

Analysis of the difference between the capital of a country and the other cities, in terms of average PM2.5 values

The analysis indicates that PM2.5 concentrations are generally elevated in capital cities, aligning with the hypothesis that densely populated areas experience higher pollution. However, a notable number of countries deviate from this pattern, displaying higher PM2.5 levels in non-capital cities. Our interactive visualization further highlights this, revealing Beijing, China, with the most significant positive difference (capital > other cities) and Thimphu, Bhutan, with the most substantial negative difference (capital < other cities).

Time Series analysis between PM2.5 and Temperature

The time series suggests a tendency for PM2.5 to be higher in colder months (autumn/winter) and lower in warmer periods (late summer/early autumn, early spring), indicating a potential inverse relationship with temperature. Temperature follows a typical seasonal pattern, decreasing into winter and increasing towards spring. While PM2.5 shows considerable short-term fluctuation, its overall levels appear to rise as temperatures fall and decline as temperatures rise. However, this observation doesn't confirm causation, and other factors likely influence PM2.5.

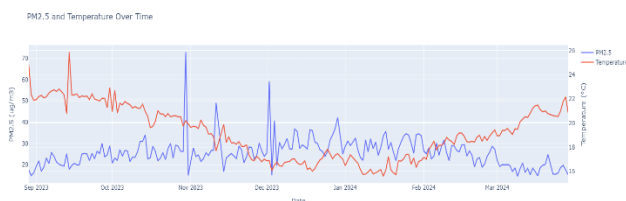


Figure 4. Time Series of Average PM2.5 and Temperature

For the individual challenge 64471, interactive dashboards were developed using Python Dash to provide insightful visualizations of the integrated data. These dashboards feature choropleth maps and an interactive earth globe to explore PM2.5 values and forest coverage, illustrating changes between 1990 and 2020 to easily identify countries with significant deforestation or forestation trends. Additionally, a global weather dashboard was created, showcasing time series of PM2.5 alongside visualizations of various weather parameters, including average temperature, precipitation, and humidity, offering a comprehensive view of the environmental context

8. CONCLUSIONS

This project has provided valuable insights into the complex relationships between environmental and demographic factors and PM2.5 levels. Our analysis of integrated global datasets revealed several key findings. Contrary to our initial hypothesis, we observed a statistically significant inverse correlation between the number of vehicles per capita and PM2.5 concentrations, suggesting that higher vehicle ownership might coincide with factors that mitigate pollution, or that other dominant sources of PM2.5 outweigh vehicular emissions in the studied regions. The influence of population was nuanced, with higher population density generally associated with increased PM2.5, but total population alone proved to be an inconsistent predictor, highlighting regional variations and the importance of considering population distribution.

Regarding forest coverage, we found a moderately weak negative correlation with PM2.5, indicating a potential, though not strongly definitive, role of forests in improving air quality. Our investigation into the pollution levels of capital cities generally supported the hypothesis that they tend to be more polluted than other cities within the same country, likely due to concentrated human activity and infrastructure, though notable exceptions underscore the importance of local context. Finally, our time series analysis suggested a potential inverse relationship between temperature and PM2.5 levels, with higher PM2.5 concentrations observed in colder months, possibly due to increased heating activities and altered atmospheric conditions.

It is important to acknowledge the limitations of this study, including potential data inconsistencies across different sources and the inherent challenges in establishing direct causal relationships from observational data. Future research could benefit from more granular, localized data and the application of causal inference techniques to further elucidate the intricate dynamics influencing PM2.5 levels. Nevertheless, our findings contribute to a broader understanding of the factors associated with air pollution and can inform policy decisions aimed at improving air quality and public health

9. REFERENCES

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Annex

Annex 1

Traffic Index List
Location, Rate[per 1000], Vehicles, Population, Year

Forest Cover Rate
Country Name Country Code Capital Continent Area (km ²) Population Density (per km ²) Population Growth Rate World Population Percentage Population Rank Forest Area 2010 Forest Area 2011 Forest Area 2012 Forest Area 2013 Forest Area 2014 Forest Area 2015 Forest Area 2016 Forest Area 2017 Forest Area 2018 Forest Area 2019 Forest Area 2020

Global Air Quality[PM2.5] - Country
Region, Year, Population-Weighted PM2.5 [ug/m3], Geographic-Mean PM2.5[ug/m3], Population Coverage [%], Geographic Coverage [%], Total Population [million people], % pop >= 5 ug/m3 [%], % pop >= 10 ug/m3 [%], % pop >= 15 ug/m3 [%], % pop >= 25 ug/m3 [%], % pop >= 30 ug/m3 [%], % pop >= 35 ug/m3 [%], % pop >= 40 ug/m3 [%], % pop >= 45 ug/m3 [%], % pop >= 50 ug/m3 [%], % pop >= 55 ug/m3 [%], % pop >= 60 ug/m3 [%]

Global Air Quality[PM2.5] - City
Region iso3 Country City/Town Year Annual mean, ug/m3 Temporal coverage note on converted PM2.5 Number and type of monitoring stations Reference for air quality Database version (year) status

GlobalWeatherRepository
country, location_name, latitude, longitude, timezone, last_updated_epoch, last_updated, temperature_celsius, temperature_fahrenheit, condition_text, wind_mph, wind_kph, wind_degree, wind_direction, pressure_mb, pressure_in, precip_mm, precip_in, humidity, cloud, feels_like_celsius, feels_like_fahrenheit, visibility_km, visibility_miles, uv_index, gust_mph, gust_kph, air_quality_Carbon_Monoxide, air_quality_Ozone, air_quality_Nitrogen_dioxide, air_quality_Sulphur_dioxide, air_quality_PM2.5, air_quality_PM10, air_quality_us-epa-index, air_quality_gb-defra-index,sunrise, sunset, moonrise, moonset, moon_phase, moon_illumination

Annex 2

Traffic Index List
Location
Rate[per 1000]
Vehicles
Year

Global Air Quality[PM2.5] - Country
Region
Year
Population-Weighted PM2.5 [ug/m3]
Geographic mean PM2.5[ug/m3]
Population Coverage [%]
Geographic Coverage [%]
Total Population [million people]
% pop >= 25 ug/m3 [%]

GlobalWeatherRepository
country
location_name
latitude
longitude
last_updated
temperature_celsius
precip_mm
humidity
PM2.5

Forest Cover Rate
Country Name
Capital
Continent
Area (km ²)
Population Density (per km ²)
Percentage
Population Rank
Forest Area 2010
Forest Area 2011
Forest Area 2012
Forest Area 2013
Forest Area 2014
Forest Area 2015
Forest Area 2016
Forest Area 2017
Forest Area 2018
Forest Area 2019
Forest Area 2020

Global Air Quality[PM2.5] - City
Region
Country
City/Town
Year
Annual mean, ug/m3

Annex 3

Description of Data Structures

1. Traffic index Table

The Traffic index Table refers to the total count of motor vehicles that have been officially registered with the appropriate government or transportation authorities within a country, region, or city. This includes passenger cars, motorcycles, buses, trucks, and other motorized vehicles that are legally permitted to operate on public roads.

field	Description	Type of data	Example
Location	Name of the country	Text	Lisbon
Rate[per 1000]	Vehicles per 1,000 people	Numeric	1059
Vehicles	Total number of motor vehicles in a country	Numeric	158658
Year	Reporting year	Numeric	2018

2. Forest Cover Rate

Location Table

The Location Table contains information about the location of the corresponding forest cover rate data. It has attributes such as country, capital, continent, longitude and area. This data will be important for easily identifying a location.

Field	Description	Data Type	Example
country_name	Name of the country	Text	Brazil
capital	Capital of the country	Text	Brasília
continent	Continent of the country	Text	South America
area_km2	Area of the country in square kilometers	Numeric	8515767

Population Table

The Population Table contains information about the population of the corresponding country in the dataset. It has attributes such as density, world population percentage and global rank.

Field	Description	Data Type	Example
population_density	Population density per km ²	Numeric	25.2841
population_percentage	World Population Percentage	Numeric	2.7
population_rank	Population rank	Numeric	7

Forest Coverage Table

The Forest Coverage Table holds information regarding the percentage of forest areas in each corresponding country during ten consecutive years. It has attributes such as the percentage of forest area and the year when the percentage was registered.

Field	Description	Data Type	Example
forest_area	Percentage of forest area	Numeric	61.20748157
year	The year when the data was collected	Numeric	2010

3. Global Air Quality[PM2.5] – Country

The Global Air Quality [PM2.5] Table contains information about PM2.5 pollution levels and population exposure in the corresponding countries. It includes attributes such as population-weighted PM2.5, geographic mean PM2.5, population coverage and total population.

Location Table: The Location table provides details about the country and the year for which the data is recorded, including the name of the region (country) and the specific year of the data.

Field	Description	Type of date	Example
Region	Name of the country	Text	Portugal
Year	Year of the data	Numeric	1998

Air Quality Information - Country Table: The Air Quality Information table includes data on PM2.5 pollution levels, such as the population-weighted concentration of fine particulate matter (PM2.5), the geographic-mean concentration of PM2.5 and the percentage of the population exposed to high concentrations of PM2.5.

Field	Description	Type of date	Example
Population-Weighted PM2.5 [ug/m3]	Avg conc. of fine particulate matter (PM2.5) that the population is exposed to weighted by population density	Numeric	13.6
PM2.5[ug/m3]	Denotes the mean annual concentration of PM2.5 in the region, without considering population	Numeric	11

	distribution		
% pop >= 25 ug/m3[%]	Specifies the percentage of the population exposed to PM2.5 concentrations exceeding 25 micrograms per cubic meter	Numeric	19.4

Population and Coverage Information Table: The Population and Coverage Information table contains data on the total population of the country, as well as the percentage of the population included in the exposure assessment and the geographic area covered by the PM2.5 monitoring.

Field	Description	Type of date	Example
Population Coverage[%]	Indicates the percentage of the region's population included in the PM2.5 exposure assessment.	Numeric	100
Geographic Coverage [%]	Reflects the proportion of the region's total land area covered by the PM2.5 monitoring or estimation.	Numeric	100
Total Population [million people]	Total population of the country	Numeric	10.124

4. Global Air Quality[PM2.5] – City

Location Table

The Location Table contains information about the location of the registered annual mean values of the PM2.5 concentration in the atmosphere within the Global Air Quality[PM2.5] - City table. It has attributes such as region, country and city name where the data was collected.

Field	Description	Data Type	Example
region	World region of the country	Text	Europe (LMIC)
country	Country of the city	Text	Romania
city_town	Name of the city or town	Text	Cluj-Napoca

Air Quality Information - city Table

The Air Quality Information - city Table contains information about each annual mean value of the PM2.5 concentration in the atmosphere along the year when the data was updated. It has attributes such as the year and the annual mean value of PM2.5 concentration measured in $\mu\text{g}/\text{m}^3$.

Field	Description	Data Type	Example
year	Year of the data	Numeric	2015
annual_mean_pm25	Annual mean PM2.5 concentration (µg/m³)	Numeric	17

5. Global Weather Repository

The Global Weather Repository Table contains information about the weather conditions in the capital city of each country. It includes attributes such as temperature, precipitation, humidity, and the last update timestamp.

Location Table: The Location table provides details about the country and its capital city, including the name, geographical coordinates (latitude and longitude) and the capital's location name.

Field	Description	Type of date	Example
country	Name of the country	Text	Portugal
location_name	Name of the city	Text	Lisbon
latitude	Latitude of the capital	Float	38.72
longitude	Longitude of the capital	Float	-9.13

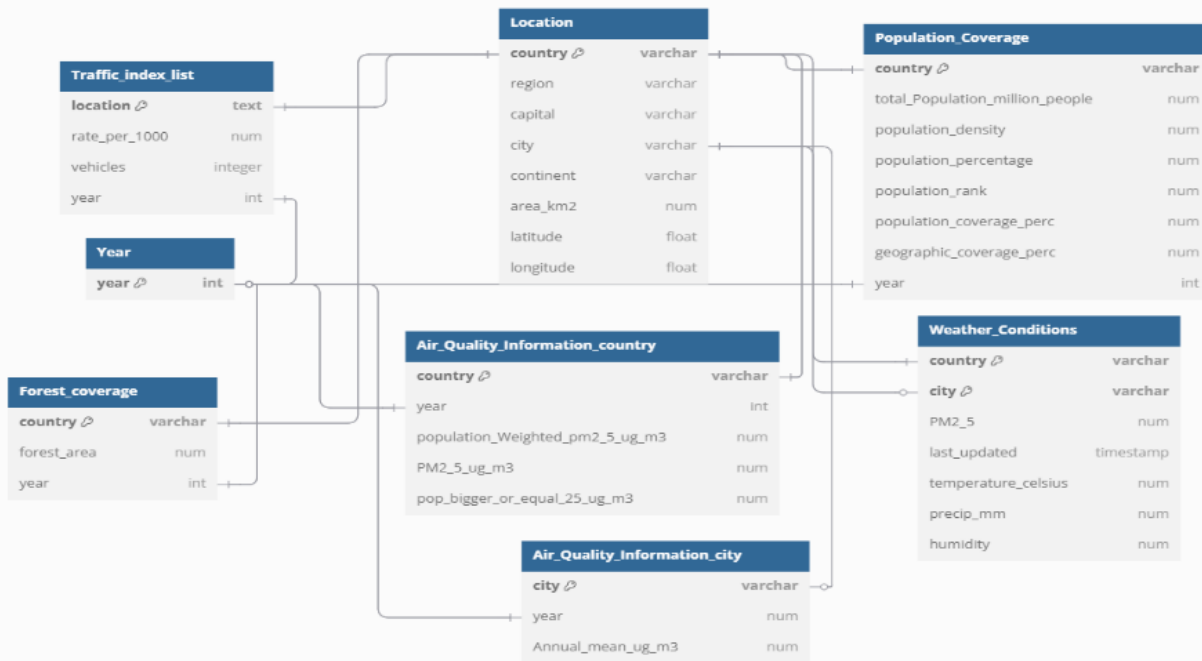
Weather Conditions: The Weather Conditions table includes data about the current weather in the capital city, such as the temperature (in Celsius), the amount of precipitation (measured in millimeters), and the humidity level in the capital. The Data Update Information contains the timestamp of when the weather data was last updated, providing context on the freshness of the information.

Field	Description	Type of date	Example
temperature_celsius	Temperature of the capital	Number	21
precip_mm	Refers to the amount of precipitation (rain, snow, or other forms of moisture) measured in millimeters (mm) in the capital.	Float	0.8
humidity	Humidity in the capital	Numeric	69
last_updated	Date & time when the data was last updated	Date and time	29-08-2023 10:30

Annex 4

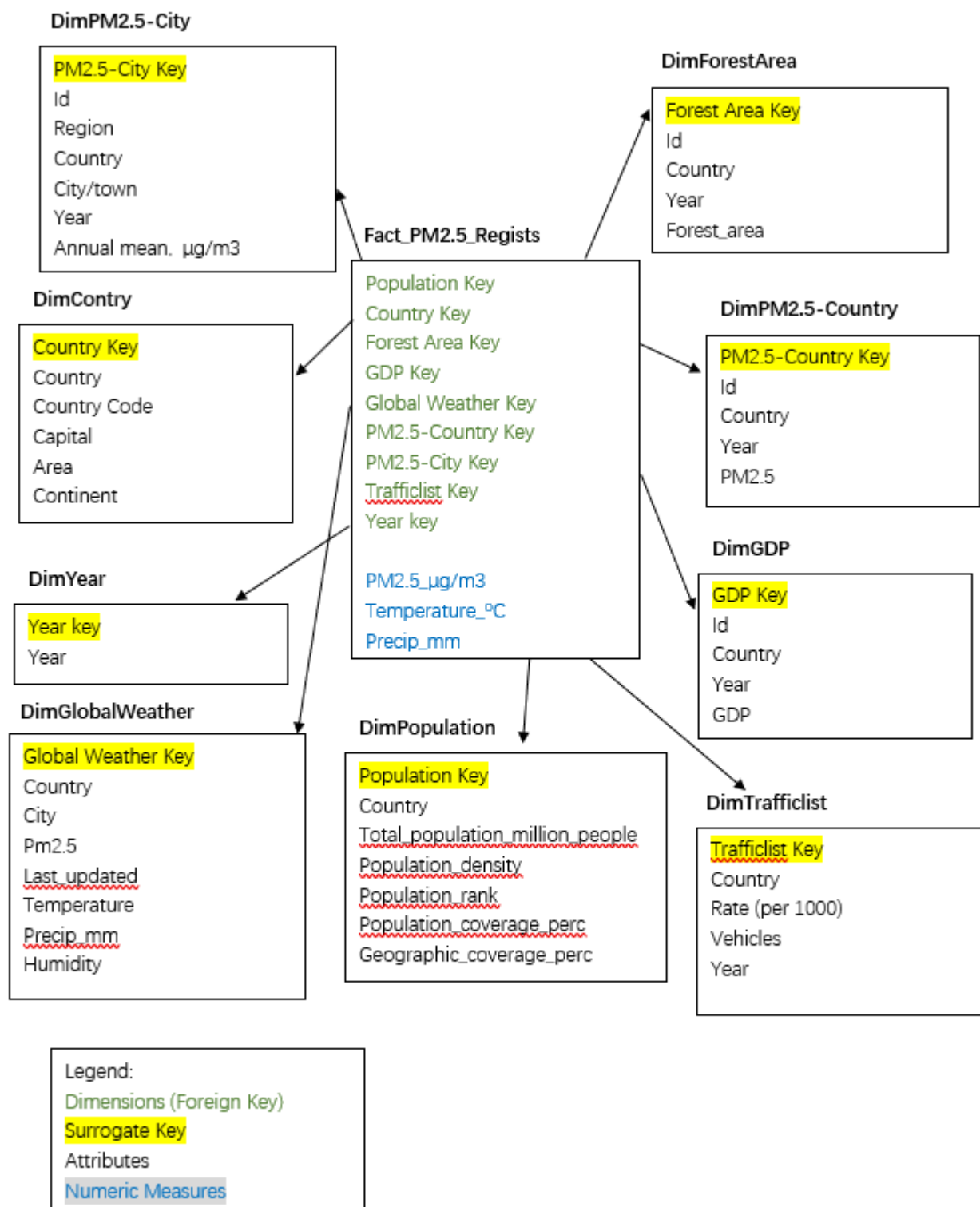
Origin Dataset	From	Target	Type Corresp.	Description
Global air quality [PM2.5]-Country	Air_quality_contry_Region	Location_Country	1:1	Name of the country
Forest Cover Rate	Forest_Country Name	Location_Country	1:1	Name of the country
Global weatherRepository	Weather_location_name	Location_City	1:1	Name of the city
Air_quality_city	Air_quality_city_City/Town	Location_City	1:1	Name of the city
Forest_coverge	Forest_coverge.ca pital	Location.capital	1:1	Name of the capital
Forest_coverge	Forest_coverge.po lulation_density	Population_covera ge.polulation_dens ity	1:1	The population density of the country
Forest_coverge	Forest_coverge.po pulation_rank	Population_covera ge.rank	1:1	The rank of the population
Global weatherRepository	Global weatherRepository .polulation_conver ge	Population_covera ge. polulation_conver ge	1:1	the percentage of the region's population included in the PM2.5 exposure assessment.
Global weatherRepository	Global weatherRepository .Total_population_ million	Population_covera ge. Total_population_ million	1:1	Total population of the country
Forest_coverge	Forest_coverge. area_km2	Location. area_km2	1:1	Area of the country
Forest_coverge	continent	Location. continent	1:1	Name of the continent
Global weatherRepository	Global weatherRepository .latitude	Location.latitude	1:1	Latitude
Global weatherRepository	Global weatherRepository .longitude	Location.longitude	1:1	Longitude
Global air quality [PM2.5]-Country	Global air quality [PM2.5]- Country.region	Location.Region	1:1	

Annex 5



Annex 6

Star Model



Annex 7

Table	How It Supports PM2.5 Analysis	Example Analytical Questions
PM2.5-Country	Core table containing yearly PM2.5 concentration levels per country	- Which countries have the highest average PM2.5?
Population_coverage	Allows normalization of PM2.5 exposure per capita; helps understand how much of the population is affected	- What proportion of the population is exposed to high PM2.5 levels?
GDP [Individual Challenge]	Enables correlation analysis between economic development and pollution levels	- Is there a link between low GDP and high PM2.5 concentrations?
Forest_coverage	Forests impact air quality; deforestation may increase PM2.5 concentration	- Do countries with lower forest coverage have worse air quality?
Trafficlist	Traffic data can help infer emission levels, especially in urban areas	- Is higher traffic density associated with higher PM2.5 values?
Country_information	Provides metadata for regional or income-level comparisons	- Do low-income regions suffer more from PM2.5 pollution?

Student – Lloyd Anthony D'Silva: 64858

Individual Challenge Report - Data Cleaning Expert

Summary:

This report consists of an extensive data cleaning initiative focused on simulated **GlobalWeatherRepository.csv** which is one of the datasets of our group project. For this individual challenge, I created a messy data and intentionally injected every kind of complex data issues. Ranging from inconsistencies to obvious mistakes to mirror the real-world challenges. A carefully designed, multi-step process was created and used to clean the data, applying advanced methods to ensure accuracy and consistency.

The Challenge: Creating Real-World Data Messiness

The foundation of this challenge involved intentionally corrupting a good dataset to create a robust data cleaning pipeline. The challenge started by taking a clean dataset and intentionally making it messy. This was done to test how well-advanced data cleaning techniques could fix real-world data problems. The messiness was based on Data Quality Framework, which looks at five areas: Accuracy, Timeliness, Completeness, Relevance and Trustworthiness.

To simulate real-world data issues, a clean dataset was intentionally corrupted in multiple ways. Hidden missing values like "-" and "N/A" were inserted into key fields, and numeric columns were polluted with text (e.g., "82F", "Freezing"). Date formats were inconsistent, coordinates were altered or swapped, and outliers such as 1000°C or logically incorrect values like PM2.5 > PM10 were introduced. Duplicate records included both exact and near matches (e.g., "NYC" vs. "New York City"), and spelling mistakes, random capitalizations, and fake entries like "Test City" were added. Dates were also tampered with, showing years far in

the future or past, making time analysis unreliable. These changes created a realistic testbed for evaluating the effectiveness of data cleaning techniques.

The Solution: A Step-by-Step Cleaning Process

To clean the dataset, a structured pipeline was followed. Hidden missing values were identified and standardized as proper nulls. Data types and formats were corrected using regular expressions, including cleaning units from numbers and fixing date strings. Missing values were imputed using the median, while categorical missing entries were removed. Outliers were detected using the IQR method and domain knowledge and corrected or removed. Duplicate entries were handled through exact match removal and fuzzy matching for near duplicates. Text fields were normalized for spelling, capitalization, and spacing. Dates were validated and corrected, keeping the most recent version where conflicts existed. Finally, irrelevant entries like "Test City" were removed to ensure a clean, consistent, and reliable dataset.

The Result: Cleaner, more Reliable Data

After cleaning, the dataset became much more reliable and usable. Below is the table summarizing *few* of the data cleaning steps.

Step	Description	Details / Counts
Initial Missing Data	Count of missing (NaN) values before cleaning	15,181
Imputation (Median Values)	Replaced missing values with median	Replaced values as follows: temperature- 22.0 precip- 0.0 humidity- 75.0 PM2.5- 7.5 PM10- 12.6 latitude- 17.25 longitude- 23.24
Missing Categorical Handling	Dropped rows with missing or "Unknown" country or location	787 ('Country'), 781 ('location_name')
Timestamp Cleaning	Dropped rows with invalid 'last_updated' timestamps	New shape: (38,249 × 10)
Outlier Detection & Capping	IQR method used to cap outliers to upper/lower bounds	temperature: 1,249; precip: 8,744; humidity: 1,159; PM2.5: 4,352; PM10: 5,173
Relational Outlier Correction	PM2.5 values exceeding PM10 were corrected	1,627 corrections
Duplicate Handling	Removed exact and near-duplicate records	63 exact; ~37,942 near-duplicates
Temporal Data Cleaning	Removed records outside valid date range (1998–2024) and kept most recent per location/day	5 removed; final shape: (228 × 11)

Individual Challenge - Rich Data Visualizations & Insights

Matei-Alexandru Lupașcu (64471)

1. Introduction

This report details the work undertaken for the individual challenge focused on creating rich data visualizations and extracting meaningful insights. The challenge involved developing interactive dashboards and visualizations to explore complex datasets related to forest cover, global weather, and air quality. The primary objective was to move beyond initial project questions and provide a more in-depth analysis through enhanced visual representation and interactive exploration.

2. Objectives

The key objectives of this challenge were to:

- Develop interactive dashboards using the Dash framework in Python to facilitate dynamic exploration of the datasets.
- Create a range of visualization types (e.g., choropleth maps, bar charts, time series plots) to represent different aspects of the data effectively.
- Incorporate user interface elements to enable interaction with the visualizations and customize their analysis.
- Extract and articulate key insights from the visualizations, providing a deeper understanding of the underlying data patterns and relationships.

3. Implementation and Results

3.1. Forest Cover Dashboard

The **forest_dashboard.py** allows interactive forest cover exploration through Year Selection, Metric Selection (percentage, change, per capita), and Map Type Selection (2D/3D). It features Top/Bottom Countries bar charts and a Time Series Analysis line chart for the top 5 countries.

Visualizations include choropleth maps (2D/3D) for spatial distribution, bar charts comparing metrics across countries, and line charts showing trends for the top 5 countries.

Key insights can be derived from this interactive tool. The choropleth maps easily help in identifying countries with high or low forest cover in any selected year. The time series analysis enables the analysis of forest cover changes over time, clearly highlighting countries experiencing deforestation or reforestation. The per capita metric provides a normalized view of forest cover relative to population density, offering a more nuanced understanding of resource availability.

3.2. Global Weather Dashboard

The **weather_dashboard.py** visualizes global weather and air quality. The user can select Metrics (temperature, humidity, precipitation, PM2.5/PM10) and a Country for time series analysis. Top/Bottom Countries bar charts and a Time Series Analysis line chart are included.

Visualizations feature bar charts comparing metrics across countries and time series line charts showing trends.

Through these features, the dashboard offers several key insights. It allows for the direct comparison of weather and air quality across different countries, highlighting regional differences. The dashboard aids in the identification of countries experiencing extreme weather conditions or facing significant air pollution challenges. Furthermore, the time series analysis reveals temporal trends and anomalies, providing valuable information for understanding changes in weather and air quality over time.

3.3. PM2.5 and Traffic Visualizations

The **pm25_traffic_visualizations.ipynb** notebook integrates a Dash application and other visualizations to analyze PM2.5 air pollution and its relationships with other factors. Key functionalities include Data Merging, demonstrating the process of combining datasets (PM2.5, forest cover, traffic) by standardizing country names to enable integrated analysis. It also features a Population-Weighted PM2.5 Time Series Dashboard, a Dash application that allows visualization and comparison of trends across multiple countries over time.

Beyond the interactive time series dashboard, the notebook uses visualizations to explore relationships within merged datasets. A heatmap displays the correlation matrix, showing linear relationships between variables like PM2.5, traffic, and forest cover. Bar plots compare PM2.5 and the number of vehicles across continents, highlighting regional differences. A choropleth map illustrates the global spatial distribution of PM2.5. A histogram visualizes the distribution of population-weighted PM2.5. These offer a multi-faceted view of PM2.5 pollution and its connections to other factors.

Insights gained from this notebook include the ability to compare PM2.5 trends across different countries via the time series dashboard, revealing variations in air quality performance. Furthermore, the data merging process facilitates the exploration of potential correlations between PM2.5 levels and other variables, such as traffic volume or forest cover, although the notebook emphasizes that further analysis is required to establish causality.

4. Conclusion

This individual challenge successfully achieved its objectives by creating rich data visualizations and interactive dashboards that enhance the exploration and understanding of environmental data. The developed tools provide valuable insights into forest cover trends, global weather patterns, and air quality dynamics. These visualizations can support informed decision-making and further research into the complex relationships between environmental variables.

Individual Challenge

Yubing Chen

Departamento Informática

FCUL-Universidade de Lisboa, Lisboa

fc52915@alunos.fc.ul.pt

Introduction

Based on the data foundation established in Phase 1 and the main research objective:

“What influences PM2.5?” I chose to work on Individual Challenge 1: Add a new question/problem that requires at least one additional data source.

For this challenge, I proposed a new question:

“Is a country’s average PM2.5 level related to its per capita GDP?”

To explore this, I introduced a new dataset titled “API_NY.GDP.PCAP.CD_DS2_en_csv_v”, which was obtained from the World Bank. This dataset contains annual per capita GDP data for countries around the world from 1960 to 2023.

I compared this new dataset with the average PM2.5 levels for each country and attempted to draw conclusions regarding potential relationships.

Integrated schema

Based on the integration schema and table structure previously completed, I added a new table named “GDP” to support the integration and linking of data (see Attachment 1).

Before the integration, I conducted data cleaning and transformation, such as:

Removing columns that were not relevant to this analysis (e.g., “Country Code”);

Renaming and harmonizing the fields to match the integrated schema.

Describing the attributes of the new table (see Attachment 2).

Analyze the data

First, based on the provided data, I created data marts using SQLite software. The data marts include three main tables: Year, GDP, and PM2.5-

Country.

a. Detail what analysis the data marts support:

1. Is a country’s average PM2.5 level related to its per capita GDP?

2. Analyzing how PM2.5 and GDP have evolved over time within individual countries

For the first question, we conducted a cross-sectional analysis by selecting ten representative countries with complete data.

We compared their per capita GDP and PM2.5 levels in 2020 and 2021 to highlight differences in development and environmental conditions.

For the second question, we used time-series line charts to visualize GDP and PM2.5 trends from 1998 to 2022 in selected countries.

This helps assess whether economic growth aligns with environmental improvement, indicating progress toward green development.

Result

To address the main question, I broke it down into two sub-questions and attached the relevant visualizations in Attachment 3 and Attachment 4.

From Attachment 3, our visual analysis shows that countries with higher GDP per capita generally have lower PM2.5 levels, while countries with lower incomes tend to experience higher air pollution, suggesting an inverse relationship between income and pollution.

From Attachment 4, we observe that in developed countries (such as Poland, Italy, and South Korea), higher GDP per capita is associated with lower PM2.5. In contrast, in some developing countries (such as Mexico, Angola, and Chile), both GDP and PM2.5 rise together. China is a special case, as it transitions from a developing to a more developed profile after 2020.